



# Substituting Inorganic Fertilizer with Fermented Oil Palm Empty Fruit Bunch Liquid Organic Fertilizer: Nutrient Uptake Efficiency and Yield of Hybrid Maize in Bengkulu, Indonesia

Suryadi \*, Usman, Elni Mutmainah, Suliasih, Rapiski, Aulia Rahma Gunawan

Department of Agrotechnology, Faculty of Agriculture, Universitas Muhammadiyah Bengkulu, Bengkulu, Indonesia.

✉ [suryadi@umb.ac.id](mailto:suryadi@umb.ac.id)

## Open Access

This article  
contributes to:



**Abstract.** Hybrid maize on acidic Ultisols depends on mineral NPK fertilizer, and its cost limits smallholder returns in Bengkulu, Indonesia. Oil palm empty fruit bunch (EFB) is an abundant local waste that can be fermented into a liquid organic fertilizer (POC). We tested how much of the recommended NPK dose POC can replace without lowering grain yield or nutrient uptake. A randomized complete block field experiment with four replications compared an unfertilized control, 100% NPK (138 kg N, 54 kg P<sub>2</sub>O<sub>5</sub>, and 60 kg K<sub>2</sub>O ha<sup>-1</sup>), 75%, 50%, and 25% of that dose combined with 3,333 L ha<sup>-1</sup> of POC, and POC alone. Full NPK produced 7.92 t ha<sup>-1</sup> of grain. Replacing 25% and 50% of the NPK dose gave 8.15 and 7.34 t ha<sup>-1</sup>, and neither differed from full NPK (Tukey HSD,  $p < 0.05$ ), whereas replacing 75% reduced yield to 6.01 t ha<sup>-1</sup>. Nitrogen uptake matched full NPK at 25% replacement and was 13% lower at 50% replacement. Agronomic efficiency of N rose from 30.6 to 37.9 kg grain per kg N at 25% replacement. Apparent N recovery (49.8 to 62.0%) was not separated by Tukey's test. Soil pH after harvest was higher than under full NPK once 50% or more of the NPK dose was replaced. Replacing up to 25% of NPK with fermented EFB liquid fertilizer maintained yield and N uptake on this Ultisol, and 50% replacement kept yield but reduced N uptake. The result comes from one season at one site and lacks a cost analysis.

**Keywords:** Acid Soil, *Zea Mays*, Liquid Organic Fertilizer, Oil Palm Waste, Nitrogen Use Efficiency.

## 1. Introduction

Ultisols cover 45.8 million ha, about a quarter of Indonesia's land area, and farmers often use them to expand agricultural production [1]. These soils are acidic, carry exchangeable aluminium, and bind applied phosphorus in aluminium and iron compounds [2]-[5]. Hybrid maize on an Ultisol therefore depends on mineral NPK fertilizer. Rising prices of non-subsidised fertilizer already burden oil palm farmers in Indonesia [1], and maize growers on the same soils buy from the same input market. A practice that cuts the mineral dose without cutting yield would lower input cost for them.

Oil palm mills supply a local source of organic matter. Empty fruit bunch (EFB) accounts for about 23% of fresh fruit bunch mass, and Indonesia produced 47.7 million t of palm oil in 2024 [6]. EFB also harbors phosphate-solubilizing microorganisms [7]. Trials with EFB products on maize have used compost or biochar. In a greenhouse study, EFB compost combined with coal fly ash raised P uptake and grain yield of maize on an Ultisol from East Kalimantan [8]. An EFB biochar-compost carrier with phosphate-solubilizing fungi improved maize growth, P uptake, and P uptake efficiency on an Ultisol from Central Kalimantan [7], and EFB compost with rice husk charcoal raised maize growth and yield on a red-yellow podzolic soil [9]. Blankson et al. [10] reported that EFB biochar and compost raised N and P use efficiency of okra above that of mineral NPK, although only the biochar-compost combination held its yield advantage into a second cycle.

### Article info

**Submitted:**  
2026-01-16

**Revised:**  
2026-03-14

**Accepted:**  
2026-04-9



This work is licensed  
under a Creative  
Commons Attribution  
4.0 International  
License.

Liquid products have received less attention. A liquid organic fertilizer enriched with phosphate-solubilizing bacteria allowed a 50% cut in P fertilizer for maize on a West Java Inceptisol [11]. Reviews of humic-based and other biostimulants list increased root growth and nutrient uptake among their most common plant responses [12], which suggests that a fermented liquid could act on nutrient uptake as well as supply nutrients.

Studies that replace mineral fertilizer with organic fertilizer in maize disagree on the best share. A meta-analysis of 133 maize studies placed the yield-optimal share of organic N at 40 to 60% [13]. In a rainfed semi-arid system in northern China, replacing 30% of chemical N raised grain yield by 9.2% and improved N recovery, while replacing 45% or 60% reduced yield [14]. A 1:1 mixture of chemical and organic N gave lower maize yield and lower N agronomic efficiency than chemical N alone in a soybean-maize intercrop [15], and 20% replacement gave the highest annual yield in a semi-arid wheat-maize rotation [16]. Other field and pot studies likewise show that moderate organic substitution can improve maize productivity and resource-use efficiency, while the response depends strongly on the replacement proportion and soil conditions [17], [18]. The single-site trials cited here ran outside humid tropical Ultisols and none used a fermented EFB liquid.

Prior work therefore shows that EFB products help maize on acidic tropical soils and that partial organic substitution can hold or raise yield. It leaves open whether a fermented liquid EFB product can take over part of the NPK dose on an Ultisol, and at what share nutrient uptake begins to fall. We addressed these questions in a field experiment in Bengkulu that graded the replacement of NPK by fermented EFB liquid fertilizer (POC, from the Indonesian pupuk organik cair) and evaluated yield together with N, P, and K uptake and N use efficiency indices [19], [20]. These indices complement fertilizer-N recovery evidence showing that management and soil organic carbon strongly influence maize nitrogen-use efficiency [21]. The reported optimum shares of 20 to 60% guided two hypotheses. H1 states that replacing up to 50% of the recommended NPK dose with POC leaves grain yield equal to that of full NPK. H2 states that replacement raises the apparent recovery efficiency of N.

## 2. Method

### 2.1 Site and soil

The experiment ran for one maize growing season on a farmer-managed field in Bengkulu Province, Sumatra, Indonesia [INSERT site name, coordinates, elevation, and planting dates]. The soil is an Ultisol. Before planting, ten cores from 0 to 20 cm were pooled and analyzed for pH (H<sub>2</sub>O, 1:2.5), organic C by the Walkley-Black method [22], total N (Kjeldahl), Bray-1 P [23], exchangeable K and Al, and cation exchange capacity. The soil was a clay loam with pH 4.62, organic C 1.38%, total N 0.11%, Bray-1 P 5.8 mg kg<sup>-1</sup>, exchangeable K 0.21 cmol<sub>c</sub> kg<sup>-1</sup>, exchangeable Al 2.9 cmol<sub>c</sub> kg<sup>-1</sup>, and cation exchange capacity 12.4 cmol<sub>c</sub> kg<sup>-1</sup>.

### 2.2 Preparation of POC

We shredded 20 kg of EFB from a local palm oil mill and fermented it under anaerobic conditions in a closed drum with 100 L of water, 2 L of molasses, and 1 L of a microbial starter [INSERT product] for 21 days at ambient temperature. The liquid was filtered through a 1 mm sieve before use. The final POC had pH 5.4, electrical conductivity 6.2 dS m<sup>-1</sup>, organic C 1.8%, N 0.42%, P<sub>2</sub>O<sub>5</sub> 0.23%, and K<sub>2</sub>O 0.85% (w/v), which gives a C/N ratio of 4.3.

## 2.3 Experimental design and treatments

The experiment used a randomized complete block design with six treatments and four blocks (24 plots of 4.5 × 3.5 m). Plots received no fertilizer (control), the full recommended NPK dose, 75, 50, or 25% of that dose plus POC, or POC alone (Table 1). The recommended dose was 300 kg urea, 150 kg SP-36, and 100 kg KCl ha<sup>-1</sup>, equal to 138 kg N, 54 kg P<sub>2</sub>O<sub>5</sub>, and 60 kg K<sub>2</sub>O ha<sup>-1</sup>. All P and K, and one third of the N, went on at planting, and the remaining N went on at 30 days after planting (DAP). POC was diluted 1:1 with water and applied as a soil drench in three equal splits at 10, 25, and 40 DAP, for a total of 3,333 L ha<sup>-1</sup> of undiluted POC (50 mL per plant). This volume supplied 14.0 kg N, 7.7 kg P<sub>2</sub>O<sub>5</sub>, and 28.3 kg K<sub>2</sub>O ha<sup>-1</sup>. A commercial hybrid maize [INSERT variety] was sown at 75 × 20 cm with one plant per hill (66,667 plants ha<sup>-1</sup>), and all plots received the same weed and pest management.

**Table 1.**  
Treatments and  
total nutrient  
supply (kg ha<sup>-1</sup>)  
from mineral  
fertilizer and  
POC.

| Code | Treatment                                                          | N     | P <sub>2</sub> O <sub>5</sub> | K <sub>2</sub> O |
|------|--------------------------------------------------------------------|-------|-------------------------------|------------------|
| T0   | Control (no fertilizer)                                            | 0     | 0                             | 0                |
| T1   | 100% NPK (300 kg urea, 150 kg SP-36, 100 kg KCl ha <sup>-1</sup> ) | 138.0 | 54.0                          | 60.0             |
| T2   | 75% NPK + POC                                                      | 117.5 | 48.2                          | 73.3             |
| T3   | 50% NPK + POC                                                      | 83.0  | 34.7                          | 58.3             |
| T4   | 25% NPK + POC                                                      | 48.5  | 21.2                          | 43.3             |
| T5   | POC only (3,333 L ha <sup>-1</sup> )                               | 14.0  | 7.7                           | 28.3             |

POC supplied 14.0 kg N, 7.7 kg P<sub>2</sub>O<sub>5</sub>, and 28.3 kg K<sub>2</sub>O ha<sup>-1</sup> in T2 to T5. Urea, SP-36, and KCl contain 46% N, 36% P<sub>2</sub>O<sub>5</sub>, and 60% K<sub>2</sub>O.

## 2.4 Measurements and efficiency indices

Plant height (10 plants per plot) was recorded at 60 DAP and ear-leaf chlorophyll (SPAD-502 meter, 10 plants per plot) at silking. A net area of 6.75 m<sup>2</sup> per plot was harvested at 110 DAP, ears were shelled, and grain mass was adjusted to 15% moisture. Kernel weight was recorded from 100 kernels. At harvest, three plants per plot were separated into grain and stover, dried at 70 °C to constant mass, and ground. Total N was determined by the Kjeldahl method, and P and K by molybdenum-blue spectrophotometry and flame photometry after wet digestion. Uptake equals tissue concentration multiplied by dry mass, summed over grain and stover and scaled to a hectare. A composite soil sample (0 to 20 cm) from each plot after harvest was analyzed for pH and Bray-1 P.

Efficiency indices follow Cassman et al. [19] and Ladha et al. [20]. Partial factor productivity (PFPN) is grain yield divided by total N applied, agronomic efficiency (AEN) is the yield gain over the control divided by total N applied, and apparent recovery efficiency (REN) is the gain in N uptake over the control divided by total N applied, expressed as a percentage. Total N applied includes the N in POC, and each plot was compared with the control plot of its own block. We did not compute the indices for the control or for POC alone, because 14 kg N ha<sup>-1</sup> makes the ratios unstable.

## 2.5 Statistical analysis

Data were analyzed by analysis of variance for a randomized complete block design, and means were separated with Tukey's honestly significant difference (HSD) test at p < 0.05. The analysis of the efficiency indices included only the four treatments with NPK. Residual normality (Shapiro–Wilk) and variance homogeneity (Levene) were checked for grain yield, and the Pearson correlation between yield and nutrient uptake used all 24 plots. The response of grain yield to the share of the NPK dose replaced by POC (0% for full NPK to 100% for POC alone) was described with a quadratic model fitted to the 20

plots of treatments T1 to T5. We solved the model for the share at which predicted yield falls to 95% of its value at 0% replacement. Calculations were run in Python with SciPy.

### 3. Result and Discussion

#### 3.1 Grain yield and yield components

Treatment affected every measured variable ( $p < 0.001$ ) except  $RE_N$  ( $p = 0.044$ ). Residuals of grain yield met the assumptions of the analysis (Shapiro–Wilk  $W = 0.95$ ,  $p = 0.30$ ; Levene  $p = 0.66$ ). Grain yield ranged from  $3.70 \text{ t ha}^{-1}$  in the control to  $8.15 \text{ t ha}^{-1}$  with 75% NPK plus POC ( $F_{5,15} = 93.9$ ,  $CV = 6.0\%$ ; Table 3, Figure 1a). Full NPK gave  $7.92 \text{ t ha}^{-1}$ . The yields with 75% and 50% NPK plus POC equaled 102.9 and 92.7% of full NPK and shared its letter, because both fell within the HSD of  $0.86 \text{ t ha}^{-1}$ . Yield dropped to  $6.01 \text{ t ha}^{-1}$  (75.8% of full NPK) with 25% NPK plus POC, and POC alone gave  $4.69 \text{ t ha}^{-1}$ , about  $1.0 \text{ t ha}^{-1}$  above the control. Kernel weight followed the same order, except that the 50% level (32.1 g) fell below full NPK (33.9 g) although its yield did not. Plant height did not separate the four treatments that received NPK, and SPAD readings dropped below full NPK only at 25% NPK plus POC (44.0 against 51.6; Table 2).

**Table 2. Plant height, chlorophyll reading (SPAD), and total plant N, P, and K uptake of hybrid maize.**

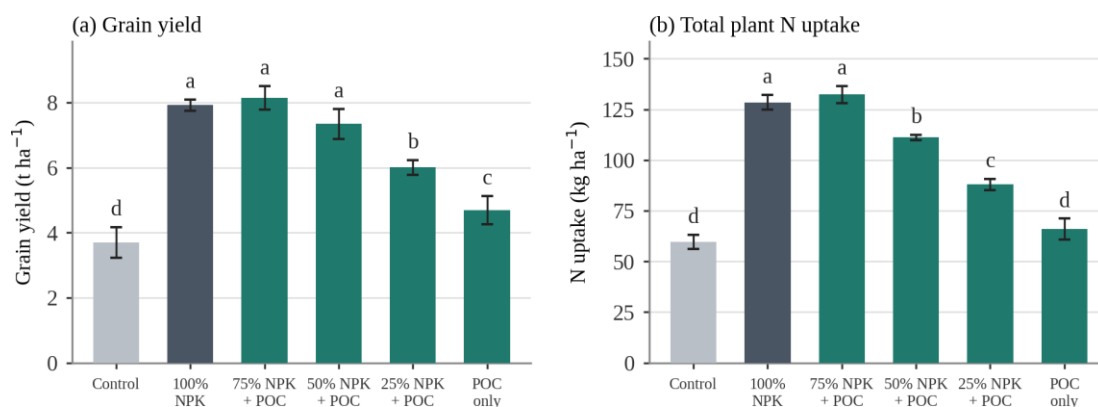
| Treatment     | Plant height at 60 DAP (cm) | SPAD at silking | N uptake (kg ha <sup>-1</sup> ) | P uptake (kg ha <sup>-1</sup> ) | K uptake (kg ha <sup>-1</sup> ) |
|---------------|-----------------------------|-----------------|---------------------------------|---------------------------------|---------------------------------|
| Control       | 149.8 ± 8.2 b               | 33.6 ± 1.6 d    | 59.7 ± 3.4 d                    | 7.6 ± 0.5 c                     | 55.8 ± 5.7 c                    |
| 100% NPK      | 217.7 ± 10.3 a              | 51.6 ± 2.1 a    | 128.4 ± 3.7 a                   | 22.3 ± 1.1 a                    | 109.1 ± 4.0 a                   |
| 75% NPK + POC | 223.8 ± 4.6 a               | 52.3 ± 1.0 a    | 132.4 ± 4.2 a                   | 24.3 ± 1.0 a                    | 119.4 ± 8.0 a                   |
| 50% NPK + POC | 207.6 ± 14.3 a              | 49.7 ± 0.3 a    | 111.2 ± 1.4 b                   | 19.7 ± 1.1 a                    | 103.8 ± 2.3 a                   |
| 25% NPK + POC | 187.1 ± 5.0 a               | 44.0 ± 2.4 b    | 87.9 ± 2.9 c                    | 14.8 ± 2.3 b                    | 85.4 ± 5.6 b                    |
| POC only      | 163.0 ± 5.6 b               | 39.1 ± 3.2 c    | 65.9 ± 5.2 d                    | 9.5 ± 0.9 c                     | 65.1 ± 0.9 c                    |

Values are means ± SD ( $n = 4$ ). Means in a column that share a letter do not differ (Tukey HSD,  $p < 0.05$ ). The F test was significant at  $p < 0.001$  for every column. DAP, days after planting.

#### 3.2 Nutrient uptake

Total N uptake was  $128.4 \text{ kg ha}^{-1}$  under full NPK and  $132.4 \text{ kg ha}^{-1}$  at 75% NPK plus POC, with no difference between them (Table 2, Figure 1b). At 50% NPK plus POC, N uptake fell to  $111.2 \text{ kg ha}^{-1}$ , which is 13% below full NPK and outside the HSD of  $8.8 \text{ kg ha}^{-1}$ , even though grain yield stayed in the top group. P and K uptake at 50% NPK plus POC ( $19.7$  and  $103.8 \text{ kg ha}^{-1}$ ) did not differ from full NPK ( $22.3$  and  $109.1 \text{ kg ha}^{-1}$ ). All three nutrients declined at 25% NPK plus POC. N uptake showed a strong correlation with grain yield across the 24 plots ( $r = 0.97$ ,  $p < 0.001$ ), and so did P uptake ( $r = 0.96$ ,  $p < 0.001$ ).

**Figure 1. Grain yield (a) and total plant N uptake (b) of hybrid maize under six fertilizer treatments. Bars are means, whiskers are SD ( $n = 4$ ), and letters mark Tukey HSD groups ( $p < 0.05$ ). Teal bars mark treatments that received POC.**



### 3.3 N use efficiency indices

PFP<sub>N</sub> rose from 57.4 kg grain per kg N under full NPK to 69.4, 88.5, and 123.9 as the NPK share fell (Table 3). AE<sub>N</sub> moved in the same direction, from 30.6 kg kg<sup>-1</sup> under full NPK to 37.9, 43.9, and 47.5, and all three POC combinations differed from full NPK. Apparent N recovery was 49.8% under full NPK and 61.8, 62.0, and 58.1% under 75, 50, and 25% NPK plus POC. The overall F test was marginal ( $F_{3,9} = 4.1$ ,  $p = 0.044$ ) and Tukey's test separated no pair of means.

**Table 3. Grain yield, kernel weight, and N use efficiency indices**

| Treatment     | Grain yield (t ha <sup>-1</sup> ) | 100-kernel weight (g) | PFP <sub>N</sub> (kg kg <sup>-1</sup> ) | AE <sub>N</sub> (kg kg <sup>-1</sup> ) | RE <sub>N</sub> (%) |
|---------------|-----------------------------------|-----------------------|-----------------------------------------|----------------------------------------|---------------------|
| Control       | 3.70 ± 0.47 d                     | 25.5 ± 0.6 d          | —                                       | —                                      | —                   |
| 100% NPK      | 7.92 ± 0.18 a                     | 33.9 ± 0.3 a          | 57.4 ± 1.3 d                            | 30.6 ± 4.4 b                           | 49.8 ± 5.1 a        |
| 75% NPK + POC | 8.15 ± 0.37 a                     | 34.6 ± 0.9 a          | 69.4 ± 3.1 c                            | 37.9 ± 6.7 a                           | 61.8 ± 2.3 a        |
| 50% NPK + POC | 7.34 ± 0.46 a                     | 32.1 ± 0.5 b          | 88.5 ± 5.5 b                            | 43.9 ± 8.8 a                           | 62.0 ± 2.9 a        |
| 25% NPK + POC | 6.01 ± 0.22 b                     | 29.8 ± 0.4 c          | 123.9 ± 4.6 a                           | 47.5 ± 5.6 a                           | 58.1 ± 13.0 a       |
| POC only      | 4.69 ± 0.43 c                     | 27.1 ± 1.3 d          | —                                       | —                                      | —                   |

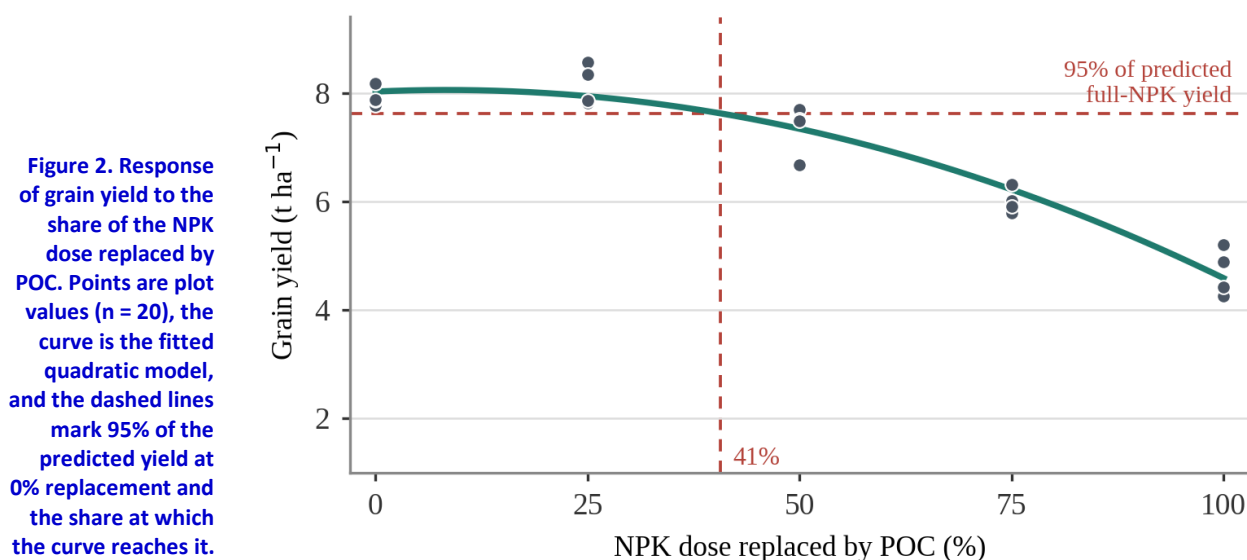
Values are means ± SD (n = 4). Means in a column that share a letter do not differ (Tukey HSD,  $p < 0.05$ ). The F test was significant at  $p < 0.001$  for every column except RE<sub>N</sub> ( $p = 0.044$ ). PFP<sub>N</sub>, partial factor productivity; AE<sub>N</sub>, agronomic efficiency; RE<sub>N</sub>, apparent recovery efficiency. Indices are based on total N applied, including N in POC, and were not computed for the control and POC alone (—).

### 3.4 Soil after harvest

Soil pH after harvest was 4.50 under full NPK (initial value 4.62) and 4.66 at 75% NPK plus POC. It reached 4.82, 4.87, and 4.94 when 50% or more of the NPK dose was replaced, and these three values exceeded that of full NPK (HSD = 0.12). Bray-1 P ranged from 7.5 to 9.8 mg kg<sup>-1</sup> across the four treatments with NPK, against 5.3 mg kg<sup>-1</sup> in the control and 5.8 mg kg<sup>-1</sup> with POC alone (HSD = 1.3).

### 3.5 Response of yield to the replacement share

A quadratic model described the response of grain yield to the replaced share x (%) of the NPK dose across treatments T1 to T5, with  $\text{yield} = 8.03 + 0.0069x - 0.000413x^2$  ( $R^2 = 0.94$ ,  $F_{2,17} = 124.5$ ,  $p < 0.001$ ). The quadratic term was significant ( $p < 0.001$ ) and the linear term was not ( $p = 0.41$ ), so yield stayed flat at low replacement shares and then declined. Predicted yield fell to 95% of its value at 0% replacement when 40.6% of the NPK dose was replaced (Figure 2).



### 3.6 Discussion

The data support H1 at 25% replacement and support it with a caveat at 50%. Replacing a quarter of the NPK dose with POC produced the same grain yield and the same N, P, and K uptake as full NPK, and the same held for yield alone at 50% replacement. This range sits inside the 20 to 60% optima reported for maize [13], [14], [16] and agrees with the pot result of Purwani and Nurjaya [24], who found the highest maize yield on a West Java Inceptisol at 50% of the recommended NPK plus a microbial biofertilizer. Our 25% level did not raise yield by the 9.2% that Wang et al. [14] recorded at 30% replacement, but the design could detect only differences above 0.86 t ha<sup>-1</sup>, about 11% of full NPK.

The 50% level needs a closer look. Its yield was 7% lower than that of full NPK, a gap the HSD could not separate, its kernel weight was lower, and its N uptake was 13% lower. Maize can hold yield while drawing on soil N, and this Ultisol holds little (0.11% total N, 1.38% organic C). The N deficit then falls on the soil pool, and repeated seasons could deplete it. Blankson et al. [10] saw such a time effect in okra, where the yield advantage of EFB amendments in the first cycle faded in the second as soil organic carbon and cation exchange capacity declined. A single-season trial cannot show whether 50% replacement is sustainable. The sole-organic treatment points the same way, since POC alone reached only 59% of full-NPK yield, in line with reports that organic inputs alone underperform in cereal systems [25].

The response at 75% NPK cannot be traced to POC nutrients alone. Total N supply at this level was 117.5 kg ha<sup>-1</sup>, 15% below full NPK, yet N uptake was equal. Three routes could account for this, and the study tested none. Biostimulant activity may have increased root growth and uptake, a common response to humic-based and related products [12]. Phosphate-solubilizing microbes carried by fermented EFB may have raised P availability, as reported for EFB materials on Ultisols [7] and for a liquid organic fertilizer on maize [11]. A higher soil pH may have eased Al toxicity and P fixation, which organic amendments achieve on acid soils [2], [3]. Organic-acid-mediated Al complexation and P mobilization provide a plausible mechanistic basis for this response [26]. Post-harvest pH under 50% or more replacement was 0.3 to 0.4 units above full NPK, and part of that gap is consistent with acidification by urea, since pH under full NPK fell 0.12 below the initial value. The 75% treatment also received 73.3 kg K<sub>2</sub>O ha<sup>-1</sup>, more than the 60 kg of full NPK, so extra K may explain part of its numerical yield advantage.

The regression places the 95% yield threshold at about 41% replacement, between the 25% and 50% treatments and inside the 30 to 60% range reported for maize [13], [14]. The model rests on five replacement levels and a single POC rate, and the POC-only treatment anchors its tail. The threshold therefore interpolates between two tested levels, and we do not offer it as a recommended rate.

POC also differs from the compost and biochar of earlier EFB trials in the organic matter it carries. Compost trials applied 5 to 20 t ha<sup>-1</sup> of solid material [8], [9], whereas 3,333 L ha<sup>-1</sup> of POC at 1.8% organic C adds about 60 kg C ha<sup>-1</sup>. The liquid cannot build soil organic matter, so any effect on the soil came through soluble constituents and microbes. That limits what a liquid product can do for an Ultisol with 1.38% organic C, and it is one more reason to expect the yield effect to depend on continued mineral inputs. AEN and PFPN rose as the NPK share fell, but part of that rise is arithmetic, since the denominator shrinks while yield holds. REN gives a cleaner test of uptake efficiency and gave only weak support to H2. Recovery was about 12 percentage points higher at 75% and 50% NPK plus POC than under full NPK, yet the overall F test was marginal and Tukey's test separated no pair. We treat H2 as unresolved. This caution is consistent with

global tracer evidence showing that N recovery is highly sensitive to soil conditions and fertilizer management [21]. Four blocks give limited power for a ratio with a CV of 9.8%.

Practical use faces a logistics constraint. The tested rate of 3,333 L ha<sup>-1</sup> requires hauling and drenching a large liquid volume, and we did not compare its cost with the saving on NPK. That comparison decides adoption and needs mill-gate prices for EFB and local prices for urea, SP-36, and KCl. The study has further limits. It covers one season, one site, and one hybrid. Roots, microbial activity, exchangeable Al, and N losses were not measured, so the mechanisms above remain hypotheses. Soil pH stayed below 5 in every treatment, so POC did not replace lime on this Ultisol. Multi-season trials with root and soil measurements and a partial budget would test both the mechanisms and the economics.

#### 4. Conclusion

On this Ultisol, replacing 25% of the recommended NPK dose with fermented EFB liquid fertilizer kept grain yield and N, P, and K uptake at the level of full NPK. Replacing 50% kept yield but lowered N uptake by 13%, and replacing 75% cut yield by about 24%. The gain of about 12 percentage points in apparent N recovery at 25 to 50% replacement was not confirmed by mean separation. We propose 25% replacement as the working rate for on-farm testing in Bengkulu, and 50% only with multi-season soil N monitoring.

#### 5. Author's Declaration

**Author contributions and responsibilities** - The authors made major contributions to the conception and design of the study. The authors took responsibility for data analysis, interpretation and discussion of results. The authors read and approved the final manuscript.

**Funding** - This research did not receive external funding.

**Availability of data and materials** - All data is available from the author.

**Competing interests** - The authors declare no competing interests.

**Did you use generative AI to write this manuscript?** - I do not use AI assistance in my manuscript.

**Declaration of generative AI and AI-assisted technologies in the writing process** - During the preparation of this work the author did not use AI to write, edit, or other things related to the manuscript.

#### 6. References

- [1] D. H. Azahari and S. Sukarman, "Impact of chemical fertilizer on soil fertility of oil palm plantations in relation to productivity and environment," *IOP Conference Series: Earth and Environmental Science*, vol. 1243, no. 1, Art. no. 012020, 2023, doi: 10.1088/1755-1315/1243/1/012020.
- [2] H. Y. Ch'ng, O. H. Ahmed, and N. M. A. Majid, "Improving phosphorus availability, nutrient uptake and dry matter production of Zea mays L. on a tropical acid soil using poultry manure biochar and pineapple leaves compost," *Experimental Agriculture*, vol. 52, no. 3, pp. 447–465, 2016, doi: 10.1017/S0014479715000204.
- [3] H. Y. Ch'ng, A. O. Haruna, N. M. N. A. Majid, and M. B. Jalloh, "Improving soil phosphorus availability and yield of Zea mays L. using biochar and compost derived from agro-industrial wastes," *Italian Journal of Agronomy*, vol. 14, no. 1, Art. no. 1107, 2019, doi: 10.4081/ija.2019.1107.
- [4] L. V. Kochian, M. A. Piñeros, J. Liu, and J. V. Magalhaes, "Plant adaptation to acid soils: The molecular basis for crop aluminum resistance," *Annual Review of Plant Biology*, vol. 66, pp. 571–598, 2015, doi: 10.1146/annurev-arplant-043014-114822.

- [5] J. V. Magalhaes, M. A. Piñeros, L. S. Maciel, and L. V. Kochian, "Emerging pleiotropic mechanisms underlying aluminum resistance and phosphorus acquisition on acidic soils," *Frontiers in Plant Science*, vol. 9, Art. no. 1420, 2018, doi: 10.3389/fpls.2018.01420.
- [6] N. S. Vinolina and C. J. Saragih, "The effect of empty fruit bunch (EFB) compost and palm oil mill effluent (POME) concentrate on pre-nursery of palm oil (*Elaeis guineensis* Jacq.) for sustainable agricultural," *IOP Conference Series: Earth and Environmental Science*, vol. 1583, no. 1, Art. no. 012002, 2026, doi: 10.1088/1755-1315/1583/1/012002.
- [7] G. I. Ichriani, S. Syehfani, Y. Nuraini, and E. Handayanto, "Formulation of biochar-compost and phosphate solubilizing fungi from oil palm empty fruit bunch to improve growth of maize in an Ultisol of Central Kalimantan," *Journal of Ecological Engineering*, vol. 19, no. 6, pp. 45–55, 2018, doi: 10.12911/22998993/92891.
- [8] F. Fahrunsyah, Z. Kusuma, B. Prasetya, and E. Handayanto, "Utilization of coal fly ash and oil palm empty fruit bunch compost to improve uptake of soil phosphorus and yield of maize grown on an Ultisol," *Journal of Ecological Engineering*, vol. 20, no. 6, pp. 36–43, 2019, doi: 10.12911/22998993/108635.
- [9] M. Mardhiana, D. Apriyani, M. Adiwena, and A. P. Pradana, "Growth and yield performance of maize at red-yellow podzolic acid soil after oil palm empty fruit bunches compost and rice husk charcoal application," *Journal of Degraded and Mining Lands Management*, vol. 8, no. 2, pp. 2653–2660, 2021, doi: 10.15243/jdmlm.2021.082.2653.
- [10] D. Blankson, E. Arthur, K. Atiah, K. A. Frimpong, P. Manfo, and S. Ravnskov, "Empty oil palm fruit bunch biochar and compost influence soil properties that drive okra (*Abelmoschus esculentus* L.) nutrient use efficiency and yield," *Soil Science Society of America Journal*, vol. 89, no. 6, 2025, doi: 10.1002/saj2.70172.
- [11] S. Sofatin, A. Citraresmini, and D. Sapta Sara, "Increasing the effectivity of liquid organic fertilizer enriched with *Enterobacter cloacae* to increase phosphate availability and yield of maize in Inceptisols," *E3S Web of Conferences*, vol. 444, Art. no. 04019, 2023, doi: 10.1051/e3sconf/202344404019.
- [12] P. Calvo, L. Nelson, and J. W. Kloepper, "Agricultural uses of plant biostimulants," *Plant and Soil*, vol. 383, nos. 1–2, pp. 3–41, 2014, doi: 10.1007/s11104-014-2131-8.
- [13] Z. Wei, H. Ying, X. Guo, M. Zhuang, Z. Cui, and F. Zhang, "Substitution of mineral fertilizer with organic fertilizer in maize systems: A meta-analysis of reduced nitrogen and carbon emissions," *Agronomy*, vol. 10, no. 8, Art. no. 1149, 2020, doi: 10.3390/agronomy10081149.
- [14] L. Wang, H. Zhou, C. Fei, et al., "Substituting partial chemical nitrogen fertilizers with organic fertilizers maintains grain yield and increases nitrogen use efficiency in maize," *Frontiers in Plant Science*, vol. 15, Art. no. 1442123, 2024, doi: 10.3389/fpls.2024.1442123.
- [15] S.-F. Lin, Y. Pi, D. Long, X. Zhu, X. Wang, J. He, and Y.-H. Zhu, "Impact of organic and chemical nitrogen fertilizers on the crop yield and fertilizer use efficiency of soybean-maize intercropping systems," *Agriculture*, vol. 12, no. 9, Art. no. 1428, 2022, doi: 10.3390/agriculture12091428.
- [16] M. Lv, M. Huang, K. Zhao, et al., "Effects of partial substitution of organic fertilizer for synthetic N fertilizer on yield and N use efficiencies in a semiarid winter wheat-summer maize rotation," *Agronomy*, vol. 13, no. 9, Art. no. 2281, 2023, doi: 10.3390/agronomy13092281.
- [17] L. Zhai, Z. Wang, Y. Zhai, et al., "Partial substitution of chemical fertilizer by organic fertilizer benefits grain yield, water use efficiency, and economic return of summer maize," *Soil and Tillage Research*, vol. 217, Art. no. 105287, 2022, doi: 10.1016/j.still.2021.105287.
- [18] H. He, M. Peng, S. Ru, Z. Hou, and J. Li, "A suitable organic fertilizer substitution ratio could improve maize yield and soil fertility with low pollution risk," *Frontiers in Plant Science*, vol. 13, Art. no. 988663, 2022, doi: 10.3389/fpls.2022.988663.
- [19] K. G. Cassman, A. Dobermann, and D. T. Walters, "Agroecosystems, nitrogen-use efficiency, and nitrogen management," *Ambio*, vol. 31, no. 2, pp. 132–140, 2002, doi: 10.1579/0044-7447-31.2.132.

- [20] J. K. Ladha, H. Pathak, T. J. Krupnik, J. Six, and C. van Kessel, "Efficiency of fertilizer nitrogen in cereal production: Retrospects and prospects," *Advances in Agronomy*, vol. 87, pp. 85–156, 2005, doi: 10.1016/S0065-2113(05)87003-8.
- [21] Z. Quan, X. Zhang, E. A. Davidson, et al., "Fates and use efficiency of nitrogen fertilizer in maize cropping systems and their responses to technologies and management practices: A global analysis on field <sup>15</sup>N tracer studies," *Earth's Future*, vol. 9, no. 5, Art. no. e2020EF001514, 2021, doi: 10.1029/2020EF001514.
- [22] A. Walkley and I. A. Black, "An examination of the Degtjareff method for determining soil organic matter, and a proposed modification of the chromic acid titration method," *Soil Science*, vol. 37, no. 1, pp. 29–38, 1934, doi: 10.1097/00010694-193401000-00003.
- [23] R. H. Bray and L. T. Kurtz, "Determination of total, organic, and available forms of phosphorus in soils," *Soil Science*, vol. 59, no. 1, pp. 39–46, 1945, doi: 10.1097/00010694-194501000-00006.
- [24] J. Purwani and N. Nurjaya, "Effectiveness of inorganic fertilizer and biofertilizer application on maize yield and fertilizer use efficiency on Inceptisol from West Java," *Journal of Tropical Soils*, vol. 25, no. 1, pp. 11–20, 2020, doi: 10.5400/jts.2020.v25i1.11-20.
- [25] Y. Liu, L. Hu, S. Zhang, Z. Yao, M. Zhou, and B. Zhu, "Organic-inorganic fertilization sustains crop yields while mitigating N<sub>2</sub>O and NO emissions in subtropical wheat-maize systems," *Agriculture*, vol. 15, no. 10, Art. no. 1108, 2025, doi: 10.3390/agriculture15101108.
- [26] Z. C. Chen and H. Liao, "Organic acid anions: An effective defensive weapon for plants against aluminum toxicity and phosphorus deficiency in acidic soils," *Journal of Genetics and Genomics*, vol. 43, no. 11, pp. 631–638, 2016, doi: 10.1016/j.jgg.2016.11.003.

**Publisher's Note** – Tinta Emas Institute stays neutral with regard to jurisdictional claims in published maps and institutional affiliations.